CSS321

Data Warehousing and Data Mining

Student Mental Health Trend Analysis

Submitted to :- Prof. SUSHITA SUSAN JOSEPH

March 2026

Team Members

S S PAVAN CHARAN (2023BCS0078)

NIKKATH ZABEEN SHAIK (2023BCS0060)

NIKHIL KUMAR (2023BCS0045)

MARCH 2026

Project Objective

Problem Statement

Student mental health is influenced by multiple academic and lifestyle factors.

The objective is to:

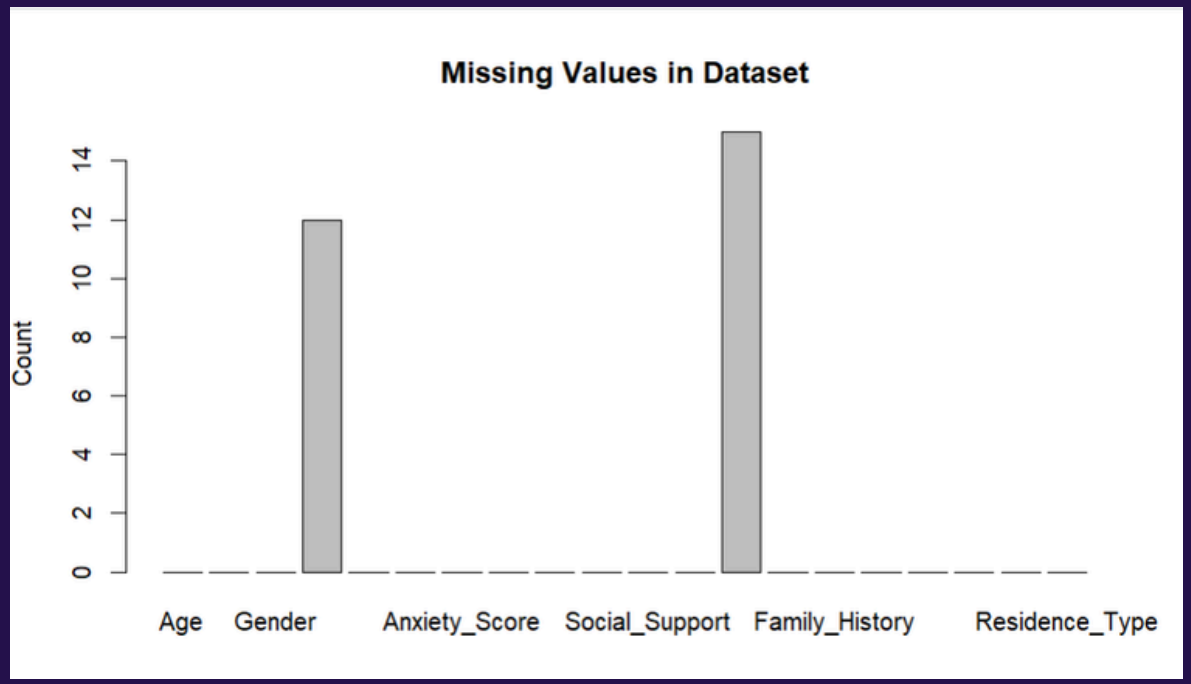
- Analyze student mental health patterns
- Build risk classification model and data warehouse
- Perform OLAP analysis
- Apply mining algorithms
- Generate insights for prediction

Dataset

- Real-world student mental health survey data
- Covers:
  - Psychological (stress, depression, anxiety)
  - Lifestyle (sleep, diet, activity)
  - Academic & financial factors
  - Multi-dimensional → suitable for analysis & prediction

Data Cleaning

- Removed duplicate records → avoids data repetition bias
- Handled missing values using median (robust to outliers)
- Converted categorical data → machine readable format
- Ensured valid and consistent dataset

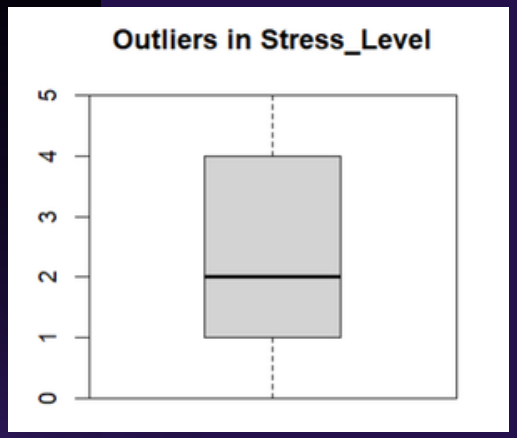


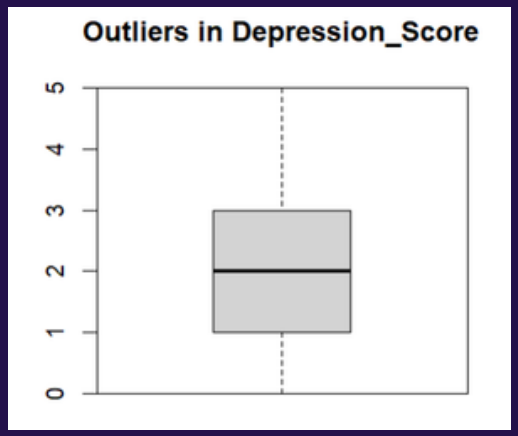
Outlier Detection

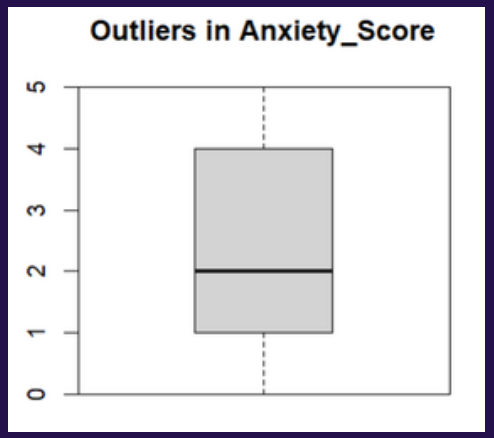
- Used IQR method to detect extreme values
- No unrealistic values in mental health scores are detected

Applied On:

- Stress Level
- Depression Score
- Anxiety Score







Smoothing & Discretization

- Converted continuous scores → Low / Medium / High
- Simplified interpretation of mental health levels
- Enabled use of rule-based mining techniques

Feature Engineering

- **Created Stress & Depression categories**
- **Risk_Category created using Stress Level, Depression Score, and Anxiety Score**
  - High → all scores ⩾ 3
  - Medium → any score ⩾ 2
  - Low → otherwise

Purpose

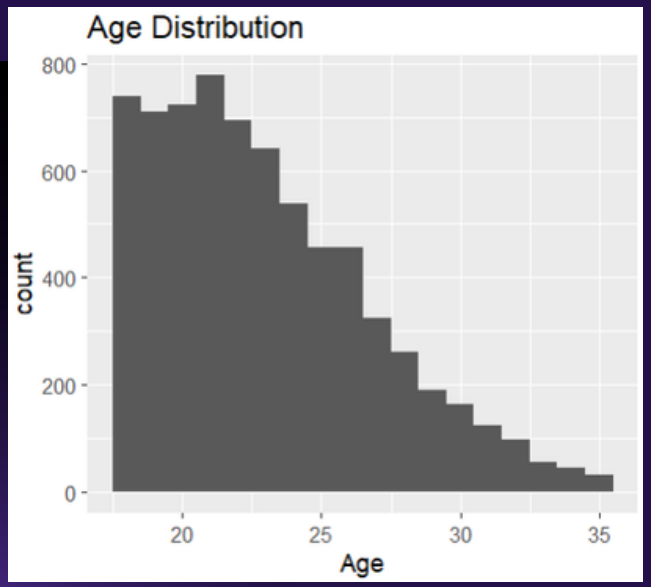
- Improves classification accuracy
- Enables OLAP summarization
- Supports Supervised learning

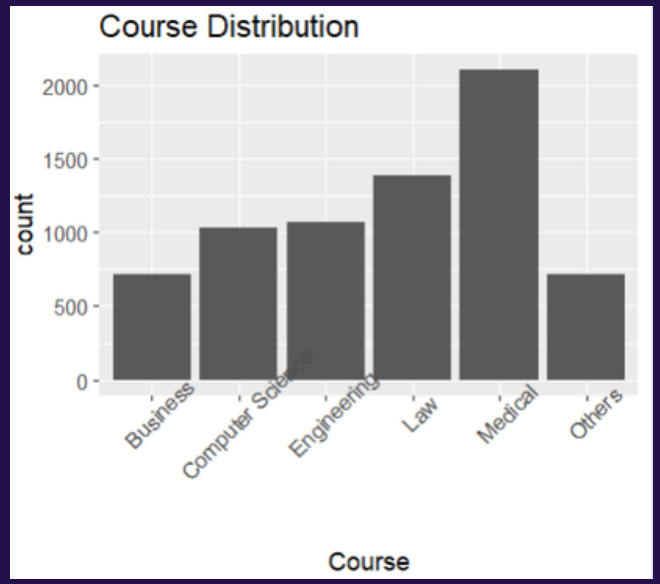
Exploratory Data Analysis

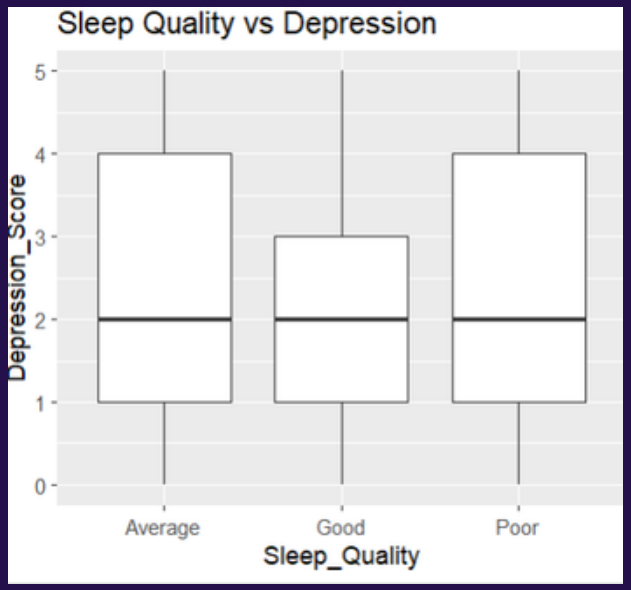
EDA Performed

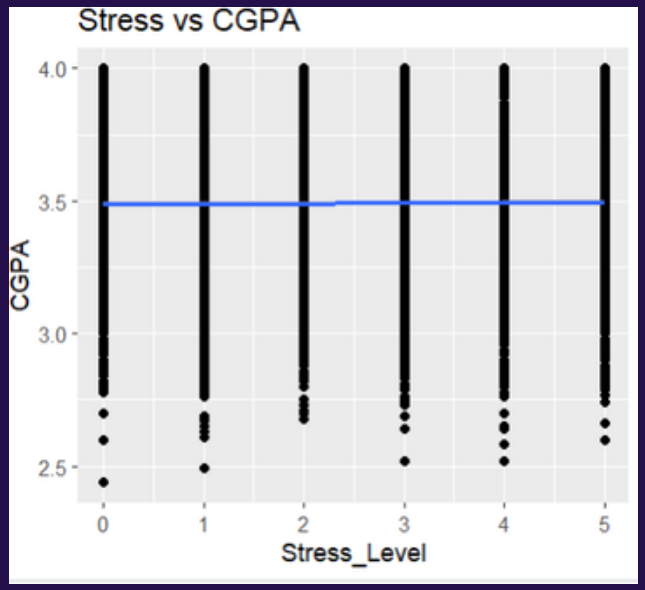
Plots:

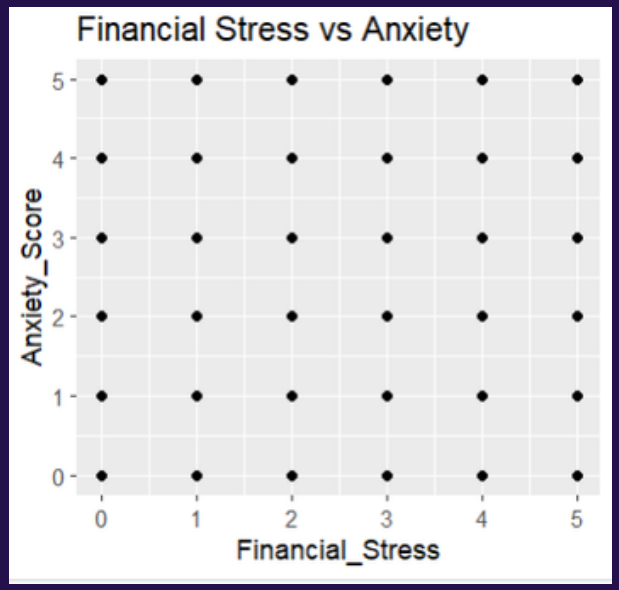
- Age Distribution
- Gender Distribution
- Course Distribution
- Stress Distribution
- Stress vs CGPA
- Sleep vs Depression



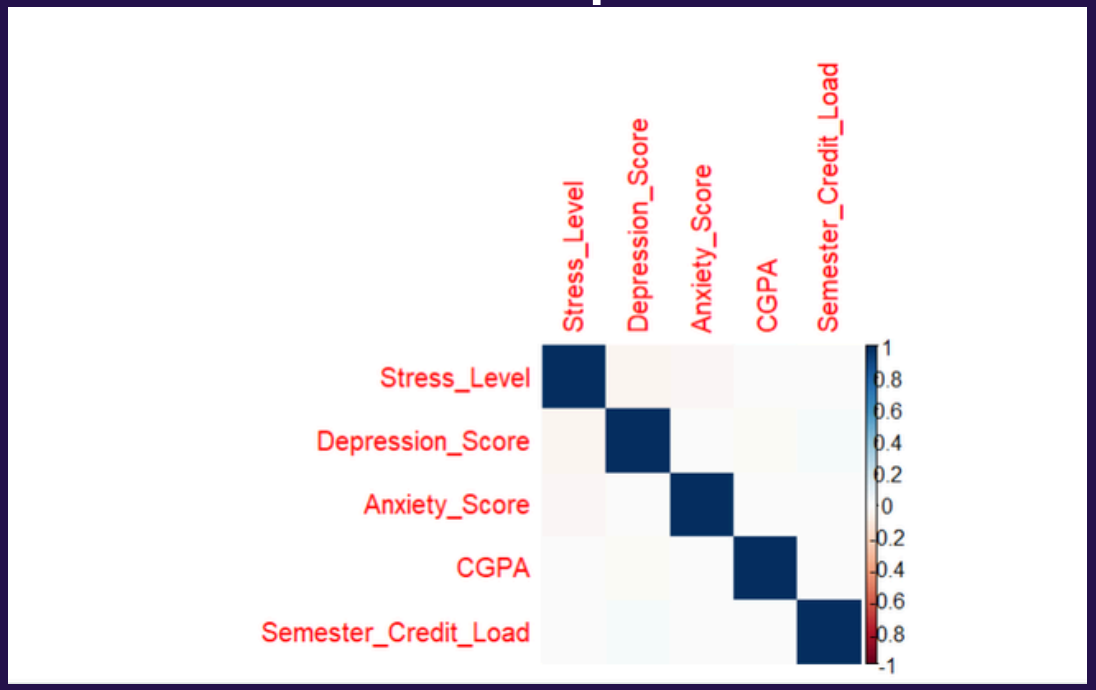








Correlation HeatMap



Variables:

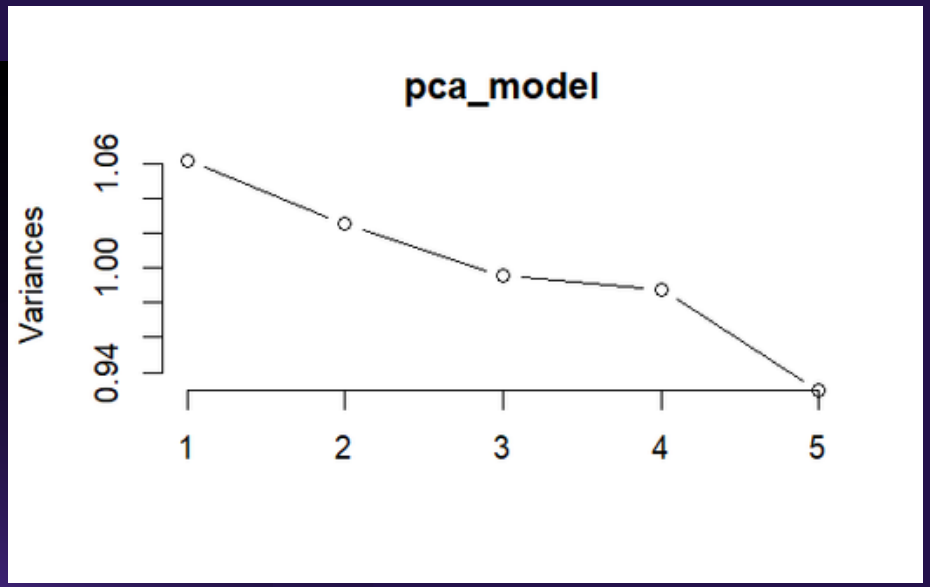
- Stress
- Depression
- Anxiety
- CGPA
- Semester Credit Load

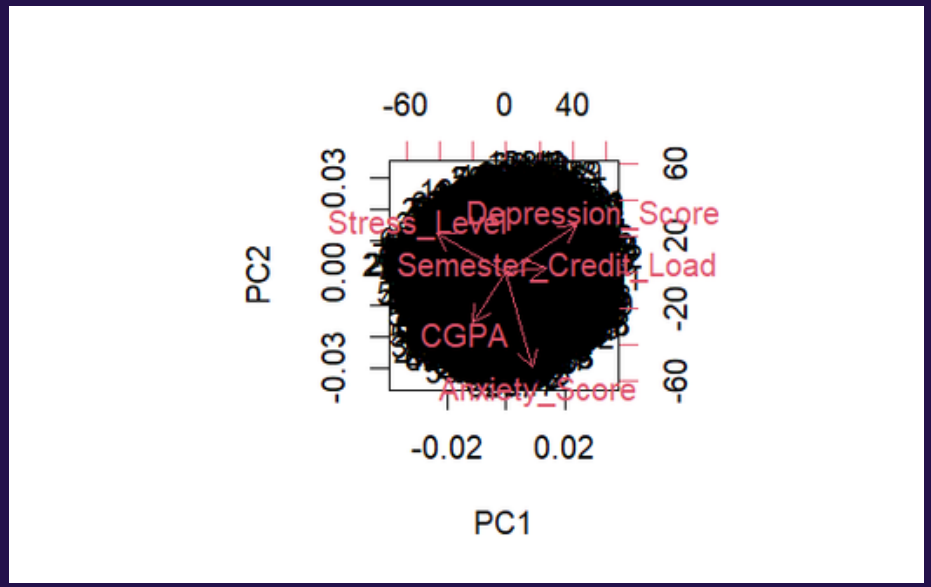
Insight:

Strong relationships identified among mental health variables.

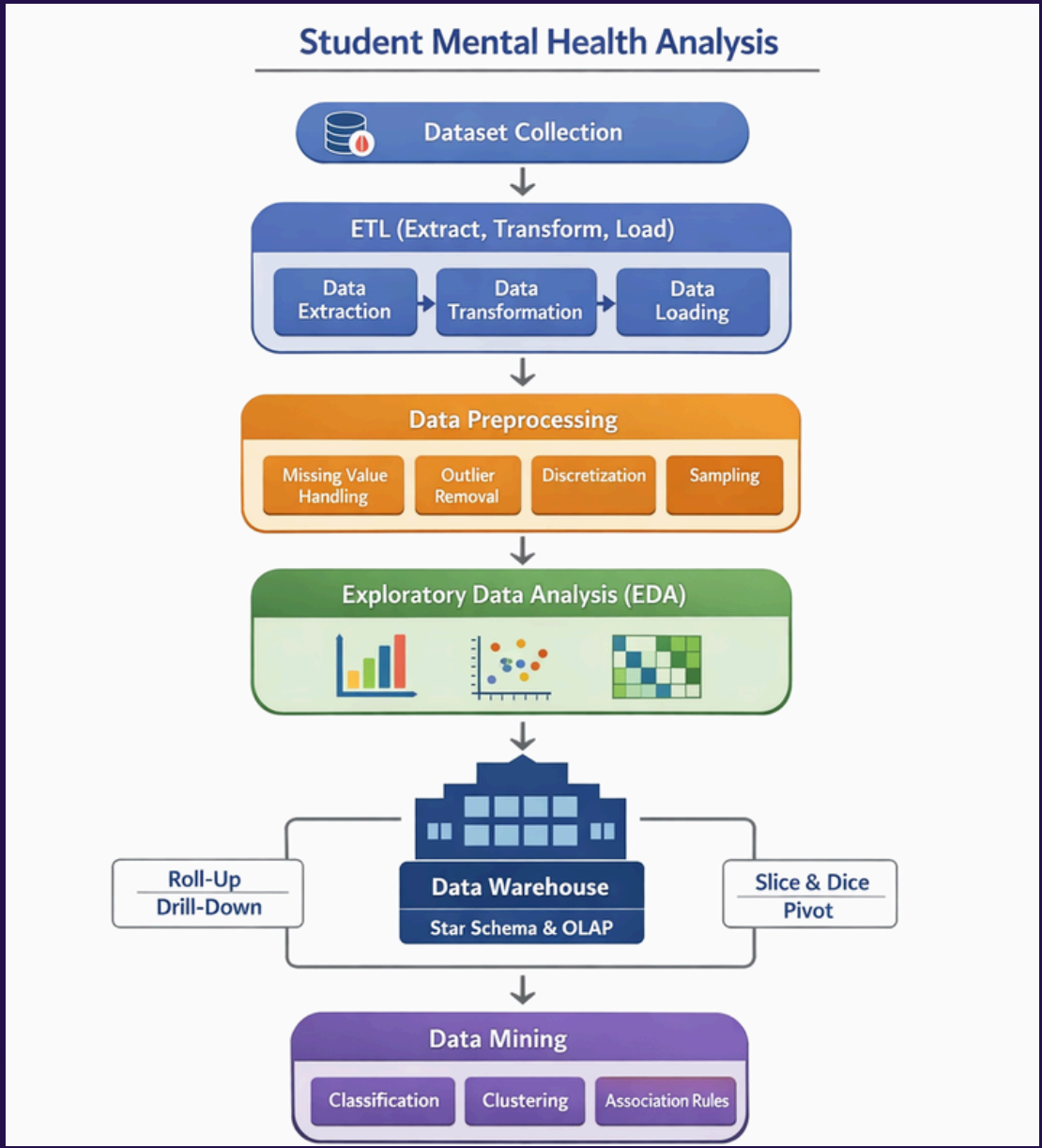
PCA (Dimensionality Reduction)

- Reduced multiple variables → principal components
- Removed redundancy among correlated features
- Captured maximum variation in fewer dimensions





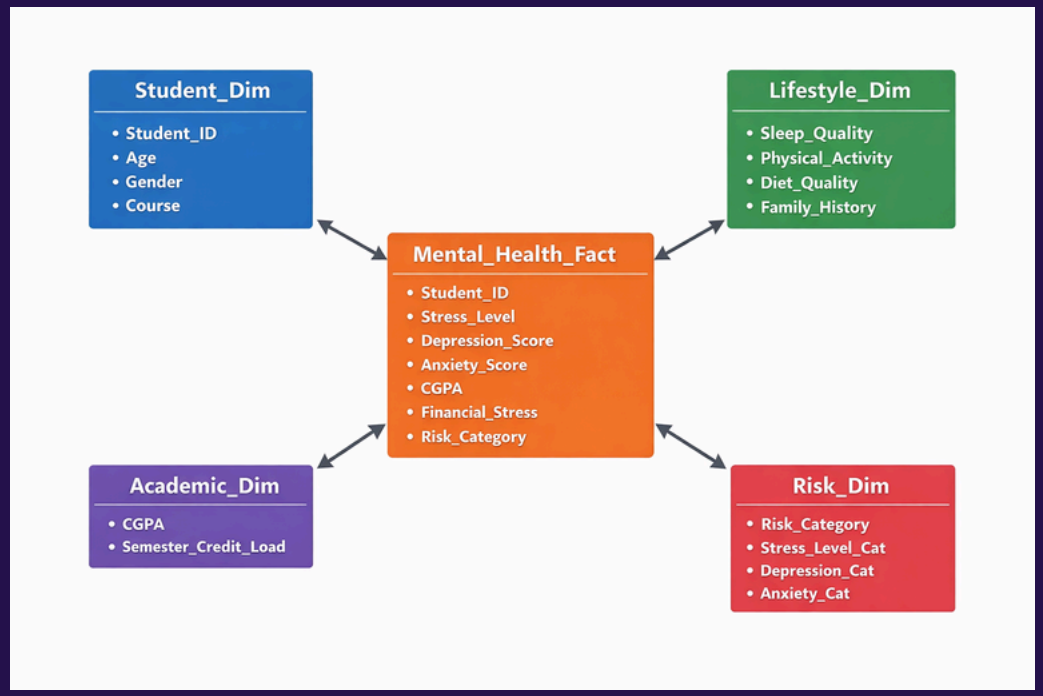
Project Flow



Data Warehouse (Star Schema)

Organized data into:

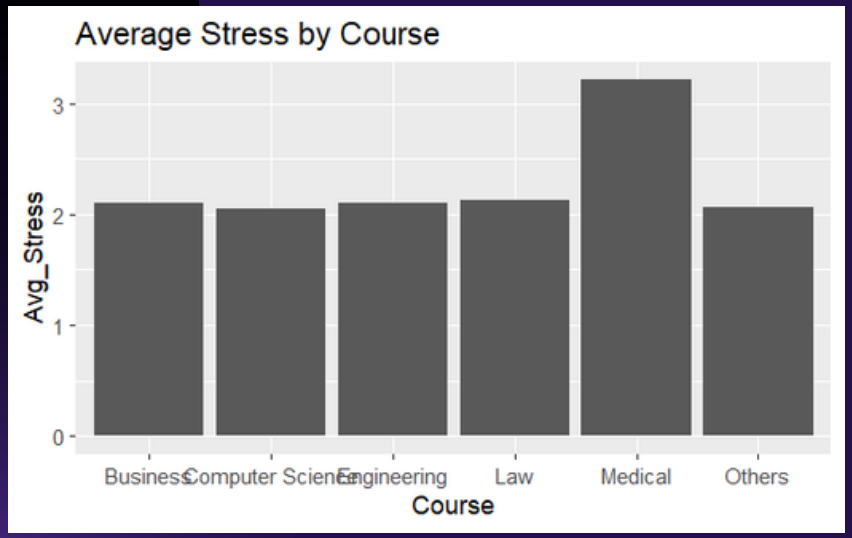
- Fact table (measures)
  - Mental health indicators
- Dimension tables (context)
  - Student Dimension
  - Lifestyle Dimension
  - Risk Dimension
  - Academic Dimension
- Structured for analytical querying
- Supports multi-dimensional analysis

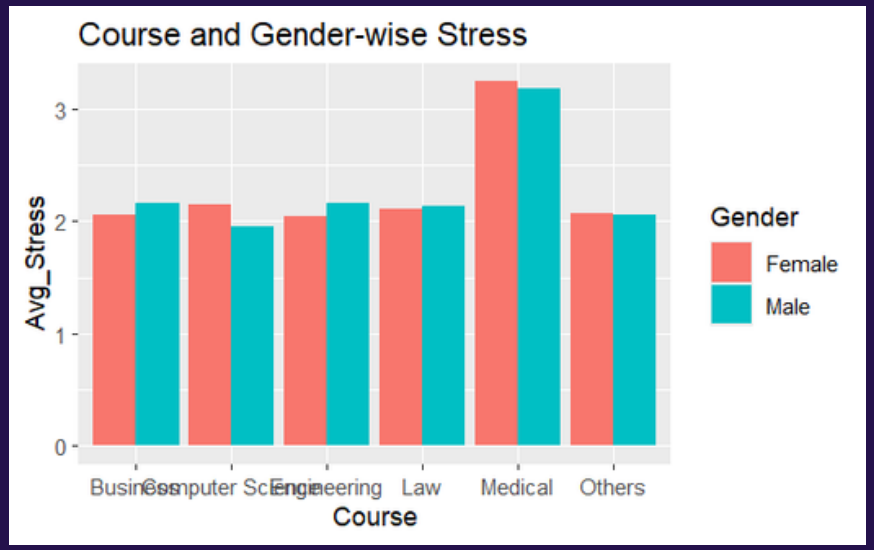


OLAP Operations

- Roll-up → summarized data
- Drill-down → detailed view
- Slice/Dice → filtered analysis
- Pivot → comparative view

Enabled data exploration from multiple perspectives





Classification Models

Built models to predict Risk_Category

Compared:

- Decision Tree (interpretable)

	Class: High	Class: Low	Class: Medium
Sensitivity	1.00000	1.00000	1.0000
Specificity	1.00000	1.00000	1.0000
Pos Pred Value	1.00000	1.00000	1.0000
Neg Pred Value	1.00000	1.00000	1.0000
Prevalence	0.09259	0.03799	0.8694
Detection Rate	0.09259	0.03799	0.8694
Detection Prevalence	0.09259	0.03799	0.8694
Balanced Accuracy	1.00000	1.00000	1.0000

- Naive Bayes (probabilistic)

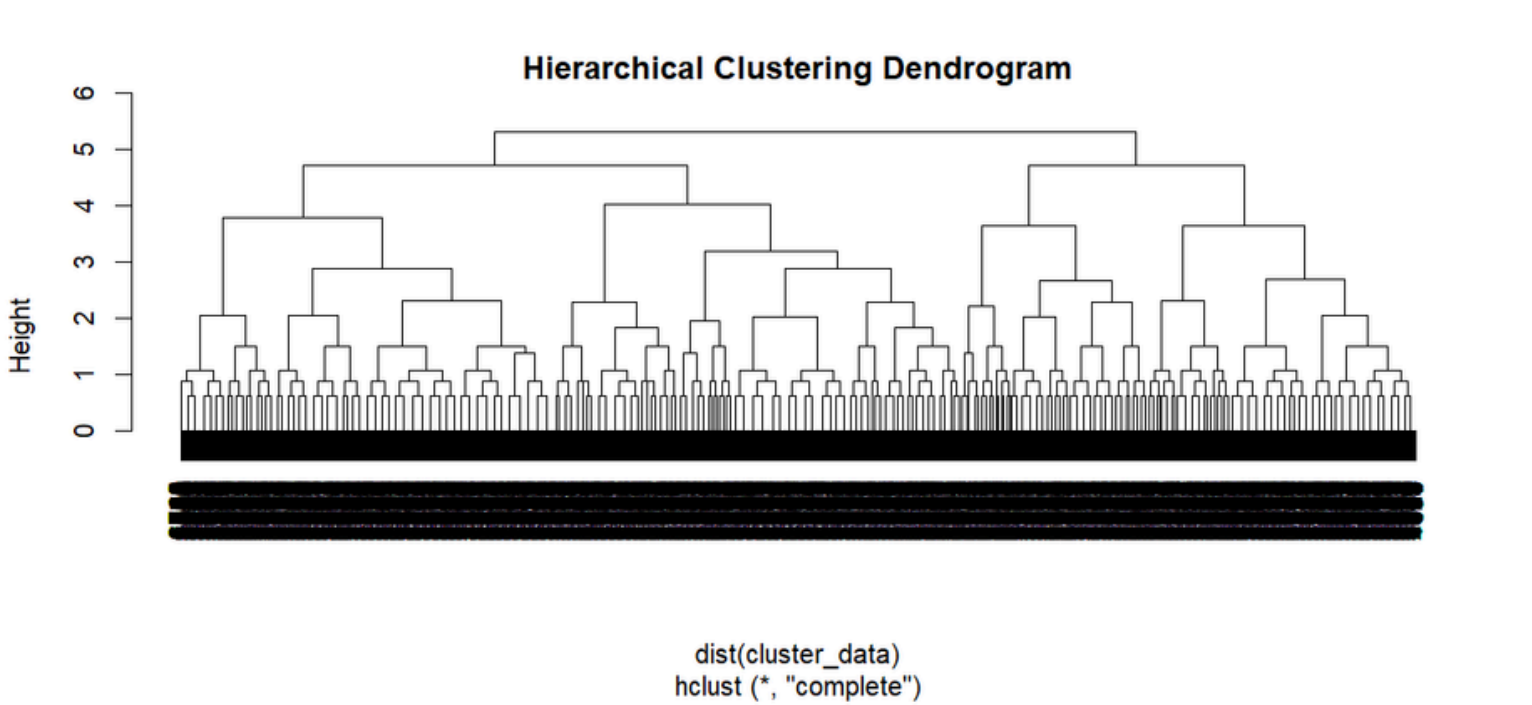
	Class: High	Class: Low	Class: Medium
Sensitivity	0.96410	0.98750	0.9984
Specificity	0.99843	1.00000	0.9709
Pos Pred Value	0.98429	1.00000	0.9956
Neg Pred Value	0.99634	0.99951	0.9889
Prevalence	0.09259	0.03799	0.8694
Detection Rate	0.08927	0.03751	0.8680
Detection Prevalence	0.09069	0.03751	0.8718
Balanced Accuracy	0.98127	0.99375	0.9846

- SVM (high accuracy)

	Class: High	Class: Low	Class: Medium
Sensitivity	1.00000	0.98750	1.0000
Specificity	1.00000	1.00000	0.9964
Pos Pred Value	1.00000	1.00000	0.9995
Neg Pred Value	1.00000	0.99951	1.0000
Prevalence	0.09259	0.03799	0.8694
Detection Rate	0.09259	0.03751	0.8694
Detection Prevalence	0.09259	0.03751	0.8699
Balanced Accuracy	1.00000	0.99375	0.9982

Clustering

- Grouped students based on mental health scores
- K-Means → fixed clusters
- Hierarchical → tree-based grouping
- Identified natural group patterns

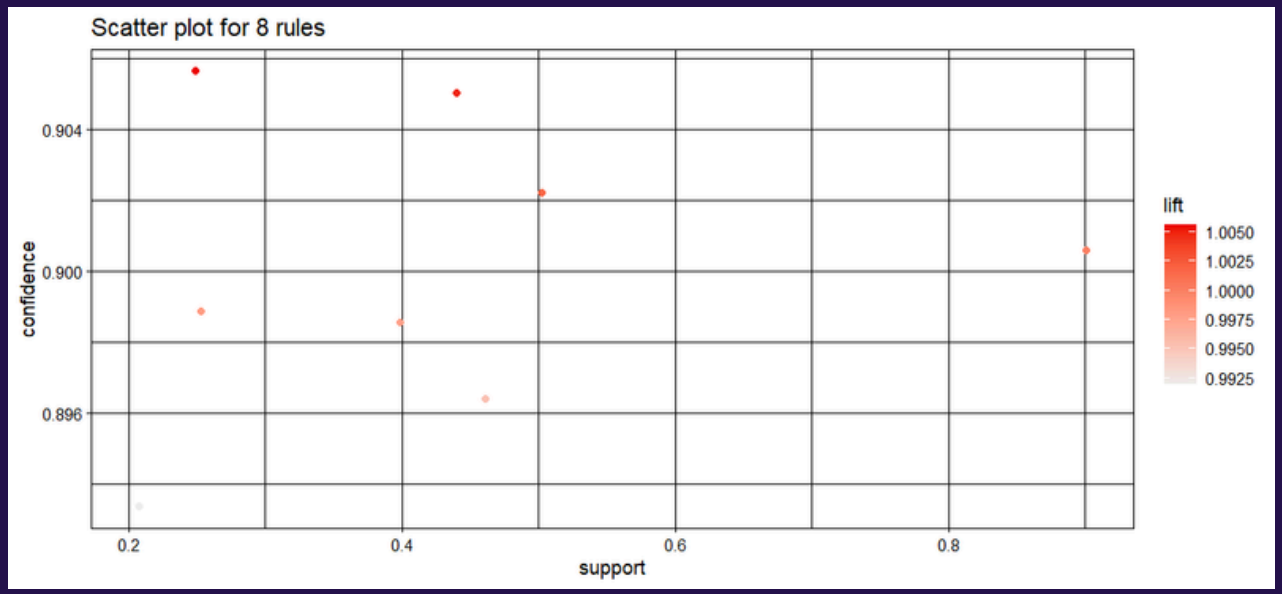


Association Rule Mining

- **Applied Apriori algorithm**
- **Rules discovered between:**
  - High Stress
  - Poor Sleep
  - High Depression

Example:

- Poor sleep → high depression
- Used support & confidence metrics



Conclusion

- Implemented complete DWDM workflow on student mental health data.
- Performed ETL, preprocessing, and warehouse design.
- Applied OLAP and mining techniques for analysis.
- Identified key patterns affecting student mental health.

Thank you!